

HURRICANE DAMAGE CLASSIFICATION

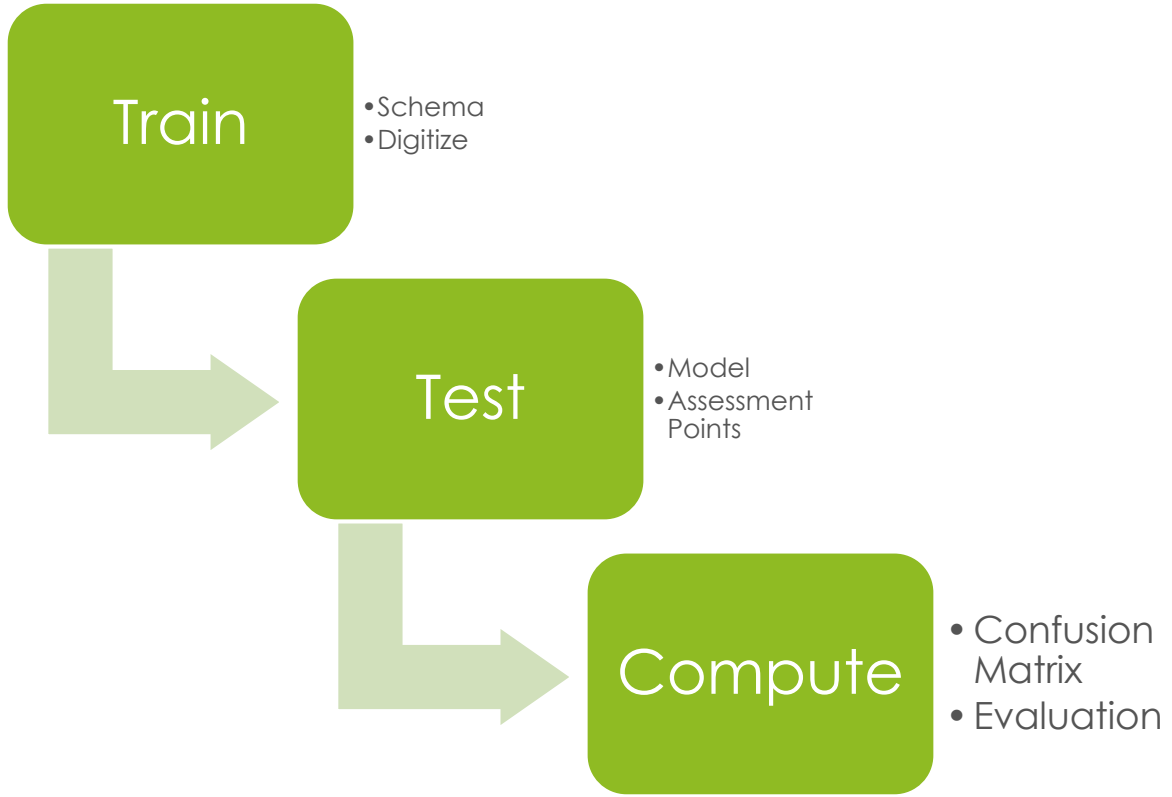
Matthew Ison

PROJECT OVERVIEW

- Develop classification model based on remote sensing imagery
  - Use data from NOAA Remote Sensing Division on Hurricane IAN
- Capture environmental and property damage
  - Use Support Vector Machine Classifier
  - Do train and test through digitized shape files
  - Score model through metrics
    - See where model failed

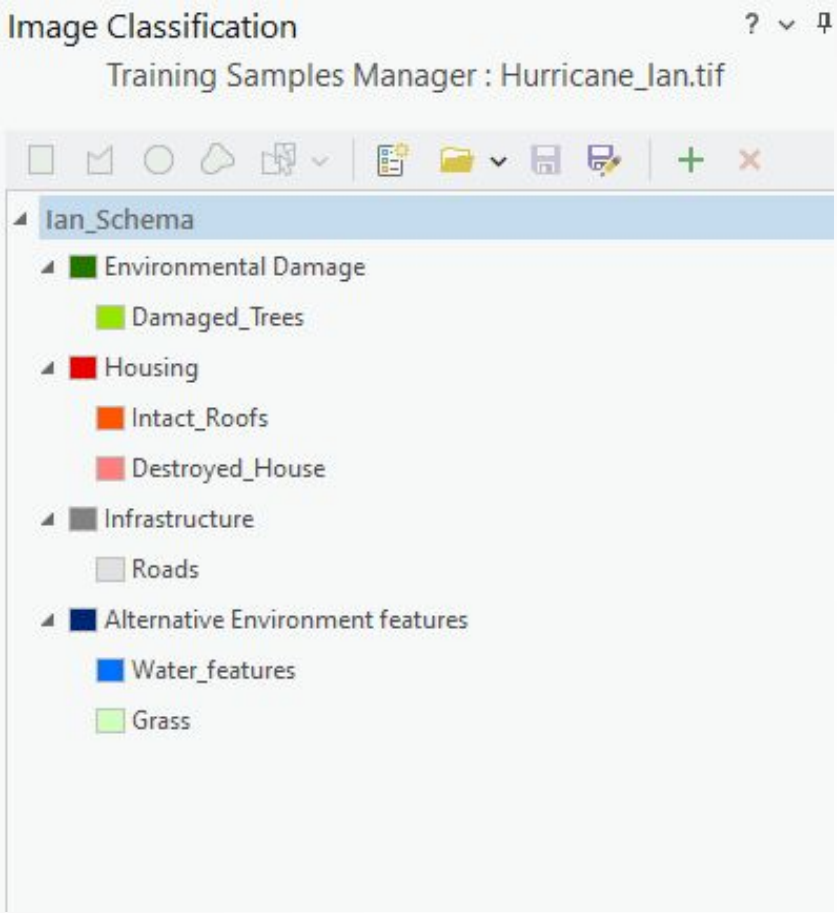


PROJECT OVERVIEW



SCHEMA

- Supervised Learning
  - Specify categories for training samples
- Provide values for categories and subcategories



CREATE TRAINING SAMPLES

CLASS TABLE

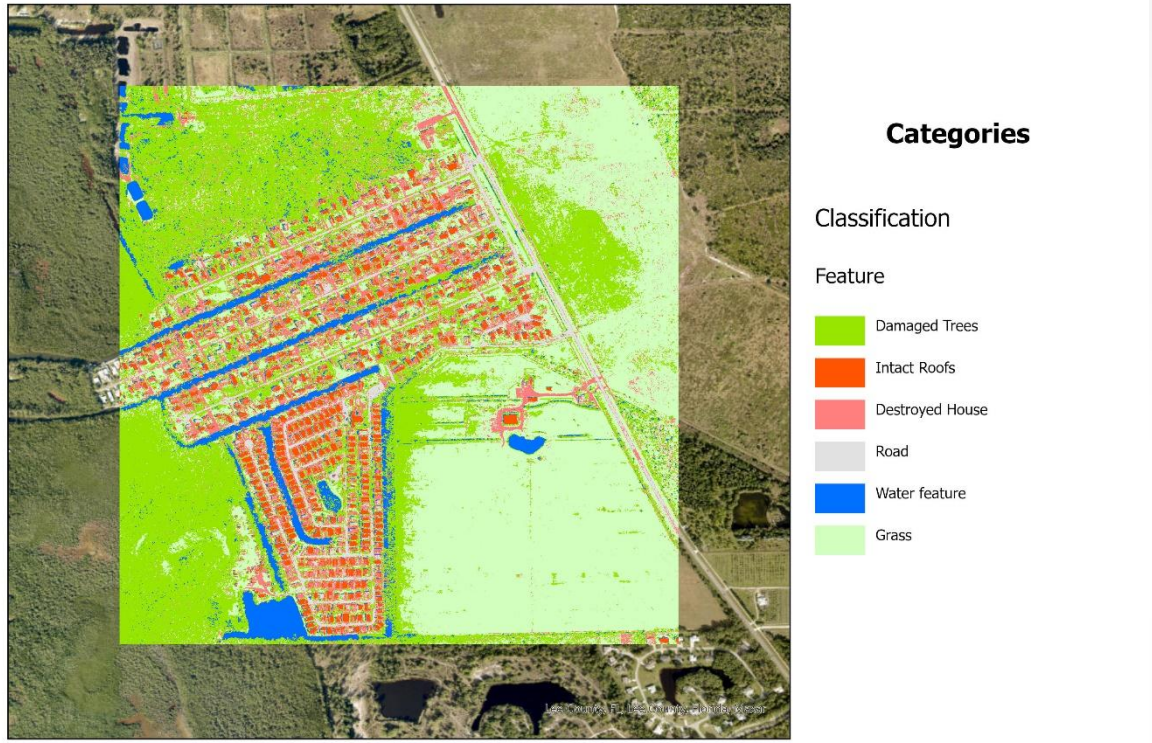
Class	# Samples	Pixels (%)
Water_features	1	0.17
Water_features	1	0.20
Water_features	1	0.06
Water_features	1	0.21
Water_features	2	2.11
Water_features	1	0.21
Water_features	1	0.39
Roads	1	0.07
Roads	1	0.09
Roads	1	0.10
Roads	1	0.04
Roads	1	0.04
Destroyed_House	1	0.11
Destroyed_House	1	0.11
Destroyed_House	1	0.11
Destroyed_House	1	0.10

DIGITIZE



MODELING

Use Image Classification Tool to classify raster using training samples (beware of runtime)



TESTING

- Create Assessment points
  - Randomly generated
  - Choose number
- Assessment based on human evaluation
  - Remember your feature's values?
  - Use base map to give more insights



CONFUSION MATRIX

- Kappa Value
  - Evaluate overall agreement between user and machine
- Model issues
  - Destroyed Houses
  - Roads

OBJECTID	Feature	Damaged Trees	Intact Roof	Destroyed House	Road	Water Feature	Grass	Total	U_Accuracy	Kappa
1	Damaged Trees	25	1	1	0	1	7	35	0.714285714	0
2	Intact Roof	0	9	1	0	0	0	10	0.9	0
3	Destroyed House	1	2	4	3	0	1	11	0.363636364	0
4	Road	0	1	2	5	0	2	10	0.5	0
5	Water Feature	0	0	0	0	8	2	10	0.8	0
6	Grass	5	0	0	0	0	33	38	0.868421053	0
7	Total	31	13	8	8	9	45	114	0	0
8	P_Accuracy	0.806451613	0.692307692	0.5	0.625	0.888888889	0.733	0	0.736842105	0
9	Kappa	0	0	0	0	0	0	0	0	0.6515

FUTURE

- Cost estimation
- Use of continuous variables in spatial prediction
- Hyper-tuning within ArcGIS
  - Integrating Python more into geospatial analysis
- Staying interested in GIS and using it in economics

